

Magnetron sputtering for development of nanostructured materials

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Abbreviations

AC	alternating current
Ag	silver
Al	aluminum
Al ₂ O ₃	Aluminum oxide
Ar	Argon
Au	gold
CD	compact disk
Co	cobalt
Co-Cr	cobalt-chromium
Co-Pt	cobalt-platinum
Cr-Ni	chromium-nickel
Cu-Ni	copper-nickel
CVD	chemical vapor deposition
DC	direct current
DVD	digital versatile disc
ECR	electron cyclotron resonance
eV	electron Volt
Fe	iron
Fe ₂ O ₃	iron(III) oxide
Ge	germanium
GHz	giga Hertz
MBE	molecular-beam epitaxy
MEMS	microelectromechanical system
MHz	megahertz
Mo	molybdenum
N ₂	nitrogen
NbN	Niobium nitride
Ni-Cr	nickel-chromium
Ni-Ti	nickel-titanium
O ₂	oxygen
RF	radio frequency
Si	silicon
Si ₃ N ₄	Silicon nitride
SiO ₂	silica
Sn	tin

TiN	Titanium nitride
TiO₂	titania
W	tungsten
WC	Tungsten carbide
WO₃	tungsten trioxide
Zn	zinc
ZnO	zinc oxide

1. Introduction

Demand on the applications of nanostructured materials in various industries is due to the unbeatable nanotechnology that provides unique characteristics retrieves from the dimensional range 1–100 nm. Nanomaterials constitute the nanostructured materials by the help of nanoparticles, nanolayers, nanofiber, nanocomposite, fullerenes, nanotube, nanowires, nanopillars, and quantum dots [1,2].

The applications of the nanostructure materials cover a wide area in chemical industries, biomedical industries, aviation industries, electronic industries, automotive industries, and household appliances (manufacturing) industries. The focus point is on how to fabricate nanostructured materials economically with respect to the application, mostly either in physical vapor deposition method or chemical vapor deposition method or hybridized method. A number of attempts has been carried out depending on the top-down approach or bottom-up approach. The top-down approach includes mechanical attrition, mechanochemical milling, lithography, laser ablation, sputtering, flame pyrolysis, electrospinning, and electrospraying [3,4].

The bottom-up approach includes thermo-chemical vapor deposition, plasma-assisted deposition, molecular beam epitaxy, sol-gel technique, solvo-thermal synthesis, chemical reduction and self-assembly techniques [5,6]. Among all the techniques, magnetron sputtering is the widespread process fully scalable for mass production and widely adopted in industries and research sectors.

2. What is magnetron sputtering?

Sputtering technique is one of a physical vapor deposition processes to develop the thin film, was pioneered in 1852 and is gained commercial success in the 1960s in microelectromechanical system industry [7]. In this process the atomic-scale materials ejected from the target materials is deposited on the exposed surface of substrate. The magnetron is generally employed as the sputtering source now a day.

Magnetron utilizes strong electric and magnetic fields and plays a dominant role to guide the plasma particle toward the target material. The powerful magnet is responsible to confine the glow discharge plasma to the region closest to the target. In magnetic field, the

electron follows a helical path around the magnetic line of force, experiences more ionizing collisions with the gaseous neutrals near the target. Here the plasma is created and the positively charged ions from the plasma are accelerated by an electrical field and strike the negative electrode (target) with a sufficient force to eject atoms. The atoms ejected from the target have a wide energy distribution up to tens of eV (100,000 K) [8].

The path of the atom from target-to-substrate depends on vacuum condition, degree of ionization, power level, gas pressure, substrate bias etc. Only a small fraction of the ejected particles are ionized in the path, i.e., up to 1%. During sputtering, 95% of incident energy goes into the target and only 5% of incident energy is carried off by target atoms (typical energy of 5–100 eV) [9].

These fractions of highly energized atom ballistically move in a straight line to deposit on the substrate. Also secondary electrons are generated from the target, accelerated away from the target and have a main role to retain the ionization in plasma. The ballistic impact of atom to the substrate can be controlled by changing the background gas pressure. In nonreactive sputtering process, generally inert Ar is taken as sputtering gas. In magnetron sputtering, if the target material is conductive, then it is connected to direct current (DC) power supply and if the target material is insulator then it is connected to radio frequency (RF) power supply. Commercially, the magnetron sputtering sources are available in a variety of geometric forms such as tubular, circular, and rectangular. The deposition rate in magnetron sputtering can be improved by maintaining the higher density of ions. It improves ionization process that leads to the increasing of the sputter yield ultimately. The sputter yield can be calculated by the ratio of the number of atoms knocked out from the target to the number of the incident atom to the target [10].

A typical magnetron sputtering process is given in Fig. 8.1. The sputter yield influenced by the incident ion energy, the total size (masses) of the ions, the binding energy of atom at the target surface, sputtering gas, sputter power, substrate temperature, substrate bias, and the angle of incident ions. Fig. 8.2A and B show the sputtering deposition process and sputtering mechanism respectively.

3. Market size of magnetron sputtering

The global magnetron sputtering system market is anticipated to increase as per the market demand with an appreciable rate during the forecast period from the year 2020 to 2027 [11]. Global sputtering targets and evaporation materials market growth rate is expected to increase in the coming years due to the increasing demands of several industries especially from electronics, automobile part, energy, optical, biomedical and other related industries. The sputtering targets and evaporation materials market revenue analysis for the year 2017 has been estimated to be \$3036 million. Again, the predicted value from the year 2018 to 2023 grows at a CAGR of 2.2% to reach up to \$3.4 billion [12].

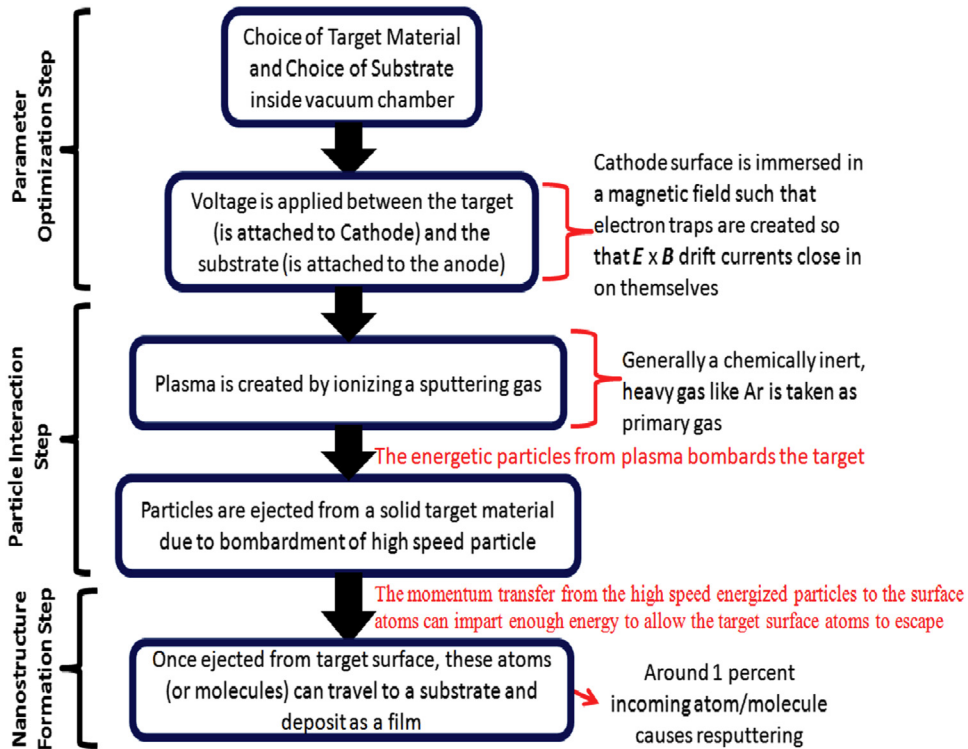


Figure 8.1

Steps involved with sputtering deposition.

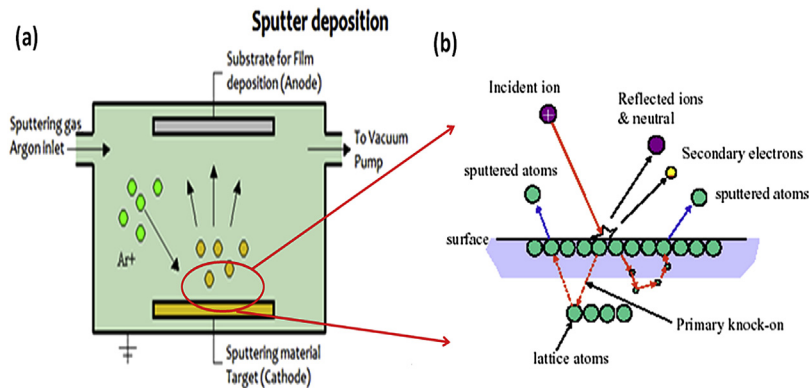


Figure 8.2

(A) Sputtering chamber at the time of deposition, (B) mechanism of deposition (sputtering/resputtering atoms).

Currently, in this year projected horizon, the market was raising at a constant rate and with various key players. The current key players in magnetron sputtering system market report are: Showa Shinku, Shincron, Torr International Inc., Von Ardenne, Buhler Leybold Optics, ULVAC, Hongda Vacuum, Angstrom Engineering, SKY Technology and Angstrom Sciences Inc. Sputtering targets and evaporation material coatings has great demand (with wide range of applications) in semiconductor components, automobile mirrors, displays, industrial tools, sensors, data storage devices, batteries, photovoltaic cells, optical instruments, decorative products, and for enhancing the properties of metal surface [13].

4. Advantages of magnetron sputtering

Various advantages are given below due to which magnetron sputtering have high demand when comparing to other deposition processes [14,15]:

- Compatible to develop the nanostructured materials using variety of materials such as metals, alloys, oxides and composite materials.
- Gives a dense film as comparison to other deposition processes.
- Extremely high adhesion of films with the substrate.
- High deposition rates.
- High-purity of the film formation.
- Possible to sputter on the heat-sensitive substrate.
- Magnetron sputtering process can operate at lower pressures.
- Can be modified the substrate surface by the reaction (reactive plasma deposition).
- Gives excellent uniformity in large-area substrates like architectural glass.
- The top-down approach performance of sputtering can be adopted in the advanced epitaxial growth process.
- The target of the sputtering process provides a stable, long-lived ionization source.
- The space adjustment between the source and the substrate is possible.

5. Magnetrons sputtering techniques in nanostructure fabrication

The use of external magnetic field is mainly differentiating the magnetron sputtering from glow discharge. The magnetic field can be developed by the application of permanent magnet or electromagnet or combination of the both. The confined magnetically plasma is responsible to sputter the atom from cathode target. There are several techniques of magnetron sputtering deposition utilized for the fabrication of nanostructured materials which are discussed below:

5.1 DC magnetron sputtering processes

Direct current magnetron sputtering is generally used for the deposition of metal, alloy and compound nanostructure. Here the magnetron is applied to deposit conducting materials by DC sputtering technique. The cost involved in this technique is lower than the RF sputtering, due to simple design and manufacturing. DC power consumption is depending on the size of the target. DC magnetron sputtering is well adopted for the fabrication of a microelectromechanical system (MEMS) due to the factors: a conductive tracks in direct contact with the Si-wafer; easy compositional control; uniform deposition throughout the cross-section [16].

5.2 RF magnetron sputtering

DC magnetron sputtering is unable to sputter the low conducting materials and insulator materials, because of nonpassing of the current through the target materials and the problem of charge accumulation. This limitation can be efficiently replaced by using RF magnetron sputtering. To overcome this stage, the alternating current with high frequency is required. In RF magnetron sputtering, the bias of anode-cathode is varied at a high rate. During sputtering at high frequency fluctuating field, the produced ion and electron have dissimilar mobilities that cover the different distances in each half cycle. The matching of impedance is a requirement in RF magnetron sputtering to ensure the maximum power is gained into the plasma.

RF magnetron sputtering has the advantages to deposit not only insulator but also conducting materials, polymer, ionic and covalent compounds to sputter away at rates depending on their specific sputtering yield. The sputter yield of ionic and covalent compound materials is much lower than the metallic materials. Dielectric materials can be deposited with sputtering yields only 10% of the metals. The power consumed for ceramic or insulator material deposition by RF is much less when compared to DC magnetron sputtering. Therefore, the deposition rate for insulator is very much lower than DC magnetron sputtering. When comparing to DC magnetron sputtering equipment, RF magnetron sputtering equipment is very complex and more expensive [17].

5.3 Ion-beam magnetron sputtering

Ion-beam sputtering process has a provision to control the energy as well as the flux of ions independently. Here the target is external to the ion source. The ion source is generated from the hot filament ionization gauge without use of any magnetic field.

Then the ionized matter is confined by using magnetic field present in the magnetron and accelerated by electric field flows from a grid toward a target. Once the ion left the source, they are neutralized by electrons from a second external filament. These neutralized atoms bombard on the target surface, which can eject the surface atoms either from the insulating or conducting targets. There is a pressure difference generated between the chamber of ion generation and the chamber of substrate. The inlet of gas is present at the source chamber. So, there is less of a chance of contamination. This type of magnetron sputtering device required more maintenance [18].

5.4 Reactive magnetron sputtering

In reactive magnetron sputtering process, the incoming atoms from the target undergo a chemical reaction due to the presence of reactive gas inside the sputtering chamber. Generally various reactive gases can be introduced such as O_2 and N_2 to develop the thin films such as oxide, nitride as well. The reactions occur between the path from target surface and the substrate surface just before deposition. The chemical composition of the film depends on the pressure of the primary inert gas and reactive gas [19].

5.5 Inductively coupled plasma-magnetron sputtering

This is an advanced version of DC/RF magnetron sputtering coupled with nonresonant induction coil. Inductively coupled plasma is used to increase the ionized discharge in the magnetron sputtering. Inductively coupled plasma coil is placed in between the cathode target and the anode substrate, parallel to the substrate. A typical inductive call operate at 13.56 MHz using a $50\ \Omega$ RF power supply. The Inductively coupled discharges are generally controlled in the pressure range 1–50 mTorr and applied power 200–1000 W results around 10^{16} – $10^{18}\ m^{-3}$ electron density. The ionized discharge increases linearly when the power and pressure increase. The ion flux and the energy of the ions are influenced by the applied rf power and the substrate bias independently. This is the main advantage over a conventional DC magnetron sputtering discharge where the ion flux and the ion bombarding energy cannot be change independently without changing the magnetic configuration [20].

5.6 Microwave amplified magnetron sputtering

Another way to increase the ionization in magnetron sputtering is to introduce a supplementary electron cyclotron resonance (ECR) in general DC magnetron sputtering. This process can generate high plasma densities (10^{17} – $10^{18}\ m^{-3}$) generally using lower working pressures (0.1–10 mTorr) with low plasma potential. The microwave generator is present at the place between the cathode and the anode for which it is possible to increase,

an electron density of the order $1 \text{ m}^{-3} \times 10^{17} \text{ m}^{-3}$. The magnetic coils are present around the chamber periphery to achieve ECR conditions. The ECR discharges are generally controlled at a microwave frequency around 2.45 GHz with a strong magnetic field. The deposition rate in microwave amplified magnetron sputtering can be increased with the microwave power and also increasing discharge gas pressure [21].

5.7 High power impulse magnetron sputtering

The maximum power in conventional DC magnetron sputtering is limited due to the thermal load on the target given by the incoming high energized positive ions. To overcome this limitation, the power is introduced in pulses with high power densities of the order of kW/cm^2 . By reducing the duty cycle (on-time divided by the cycle-time), a corresponding increase in power during the on-time can be achieved. By this process high plasma densities can be produce with the order of 10^{19} m^{-3} [22].

6. Magnetron sputtering for fabrication of NiTi smart materials

Among the advanced material systems, the smart-structural system has high demand in structural and functional applications in various industries from aerospace industry to medical industry. By the expected applications for MEMS, various techniques were tried for the fabrication of Ni–Ti thin films. From a practical point of view, magnetron sputtering results a perfect shape memory effect in thin film as that of bulk materials [23].

In smart material sputtering process, utilization of inert gas like Ar is essential for the formation of the thin film. Properties such as transformation temperatures, shape memory behavior and pseudoelasticity of the sputtered NiTi films are sensitive to target composition, target power, gas pressure, target-to-substrate distance, deposition temperature, substrate bias, substrate rotation, good configuration of target, and the deposition environment. DC and RF magnetron sputtering process is used for NiTi shape memory thin film production, in which separately, Ni is connected with the DC and Ti is connected with the RF power to maintain the sputter yield [24]. The nanostructured film composition is controlled from separate targets by confined power ratio. It is very important to limit the impurities (typically oxygen, hydrogen, and carbon) to prevent the loss of shape memory effect and to prevent brittleness and deterioration.

7. Variation of parameters in magnetron sputtering deposition

7.1 Ar gas pressure

By increasing the Ar pressure, the resultant nanostructure gives a low dense thin film with fewer structural defects. At very low working pressures, there are fewer collisions and the

adatoms arrive at the substrate surface with high energies and results atomic peening. By atomic peening, the surface atoms are pushed into the interior of the film, increasing the density with creating the compressive stresses. By gradual increase in gas pressure, the momentum of the adatom atoms decreases by higher collision, energy decreases, peening effect decreases and the compressive stresses decreases. Further increase in pressure, decreases density of the film, increases the tensile intrinsic stress of film as well. High gas pressure reduces the energy of adatom and leads to low surface mobilities. The film grows in a columnar fashion and results in low dense film [25].

7.2 Target-to-substrate distance

The target-to-substrate distance suggests the energies of adatoms coming from the target and influences the deposition efficiency, film density, structural integrity, and stress. The small target-to-substrate distances with low gas pressure produces a film of intrinsic compressive stresses which become tensile as the Ar pressure increases. This distance should be maintained in accordance with the gas pressure. Increasing of target-to-substrate distance increases the gas pressure. On the other hand, decreasing the distance lowers the gas pressure [26].

7.3 Substrate temperature

Higher substrate temperature leads to lower the cooling rate and higher diffusivities of adatoms which results in partial crystallization in the films. At low substrate temperature, the film consists of a continuous network of uniformly wide and homogeneously distributed grain boundary like structures. This can be attributed to the lack of surface migration of adatoms to form aggregates which lead to the porous boundaries. The increase in the grain size occurs at high substrate temperature. At higher substrate temperature, the surface diffusion of adatoms increase results in denser and compact film. It is also observed that few grains are more elevated than the neighboring grains along with a reduction in density of grain boundaries [27].

7.4 Substrate bias

Most of the case sputtering deposition efficiency depends on the substrate bias. Ion bombardment during the film growth process is usually achieved in the sputtering process by means of applying small negative bias voltage at the substrate end in order to improve the quality of sputter deposited films. It is expected that the substrate-bias voltage might have a considerable effect on the film crystallinity, microstructure as well as the resputtering rate from the film. It is customary to introduce bias to the substrate during

film deposition. Throughout the process, the low energy ion bombardment of the growing film may preferentially remove loosely bonded adatoms from the surface and increase their diffusion on the substrate [28].

7.5 Substrate rotation

Substrate rotation is required for uniform distribution of adatoms. The rpm depends on the velocity of an atomic particle launching toward the substrate. Substrate rotation has a significant effect on structural and mechanical properties. In recent years, the possibility to grow nearly arbitrary shaped nanostructures by oblique incidence deposition under controlled substrate rotation in a process commonly known as glancing angle deposition has gained significant attention due to the manifold of possible applications such as photonic crystals, polarizing filters, and humidity sensors or pressure sensors [29].

7.6 Degree of ionization

The ionization rate increases by increase in applied power as well as increase in pulse length. Additional equipment may be coupled in between the cathode and anode like inductive coil or microwave coil to improve the degree of ionization. The comparison suggests an even higher ion to neutral ratio for the high-induced-plasma discharge. The calculation of the ionized flux fraction has also been made by weight gain differences on a floating and a positively biased substrate [30].

7.7 Deposition rate

In the high impulse discharge magnetron sputtering, the deposition rate is generally found to be lower than in a conventional DC magnetron sputtering discharge at the same average power. The reduction in deposition rate is due to the presence of sputtered material close to the target. In this case, many of the metallic ions will be attracted back to the target surface by the cathode potential [31].

In the presence of a confining magnetic field, as in magnetron sputtering, not all the applied voltage drops over the sheath region that forms in front of the target, but a fraction of the applied voltage will penetrate the bulk plasma and create a potential gradient inside the plasma, referred to as the plasma presheath. If a sputtered atom is ionized inside this region, it must have enough kinetic energy to escape this potential in order to reach the substrate. However, if the ion is attracted back to the target it will act as a sputtering particle. A reduction in the deposition rate is then expected to occur especially for metals with a low self-sputtering yield. The deposition rate can be optimized by varying the magnetic confinement [32].

7.8 Stress

Stress in the film plays a dominant role in film/substrate adhesion. The main factors responsible for the stress evolution in nanostructured materials is due to: the thermal expansion mismatch between developed nanostructure and the substrate causes the intrinsic tensile stress; the small density difference between amorphous and crystalline such as in NiTi. A wide range of residual stress levels (either tensile or compressive) was found in the sputtered films. Large residual stress could lead to either film cracking or delamination under tension and buckling under compression [33].

8. Nanocomposite coatings by magnetron sputtering

By sputtering deposition, there are two way to fabricate the nanocomposite: one is codeposition, and second one is multilayer deposition followed by annealing. In codeposition process, several targets are exposed to plasma to deposit simultaneously on the substrate. In multilayered deposition process, only one target is exposed to plasma and all other targets are covered by the shutter. After one by one nano-scale deposition of the target, next target starts to deposit. In this way alternate nanolayer deposition happens. Once deposition finished, annealing started in vacuum to homogenize the thin film throughout the whole thickness. Vacuum is necessary to avoid oxidation. A little diffusion already happened in the thin film, due to three reasons: setting the substrate temperature during magnetron sputtering deposition; increasing of bias to the substrate; during sputtering/resputtering interaction [34].

Khan et al. have prepared Ag-Permalloy (Ni₇₆Fe₁₉Ag₅ and Ni₇₂Fe₁₈Ag₁₀) thin films using DC reactive magnetron sputtering method in 5n (99.999%) pure Ar gas atmosphere. Also, they have prepared the Ni₇₂Fe₁₈Ag₁₀ thin film in Ar + N₂ gas atmosphere (with aggregate flow rate 60 sccm and total pressure 0.55 Pa). The films have been deposited on the flexible polymer substrate, i.e., polyethylene terephthalic ester (PETE) using mosaic alloy target at room temperature. To prepare flexible soft magnetic thin films, polymer substrate is more suitable than glass substrates (brittle and nondeformable). By comparing these films preparation methods, it is observed that the reactive gas (N₂) makes changes in the chemical composition of the thin films. Further, addition of N₂ reduces the average film deposition rate. In reactive nitrogen atmosphere, average film deposition rate is 1.8 nm/min and in Ar atmosphere 2.05 nm/min [35].

SEM images and EDS analysis of the as-deposited (done with 115 W sputtering power) thin films on PETE substrate are shown in Fig. 8.3. Thin films deposited in Ar gas only are shown in Fig. 8.3A and B. Their average grain size is 25.0 and 23.2 nm respectively. The surface of the films is smoother. Fig. 8.3C shows the film deposited in Ar + N₂ gas

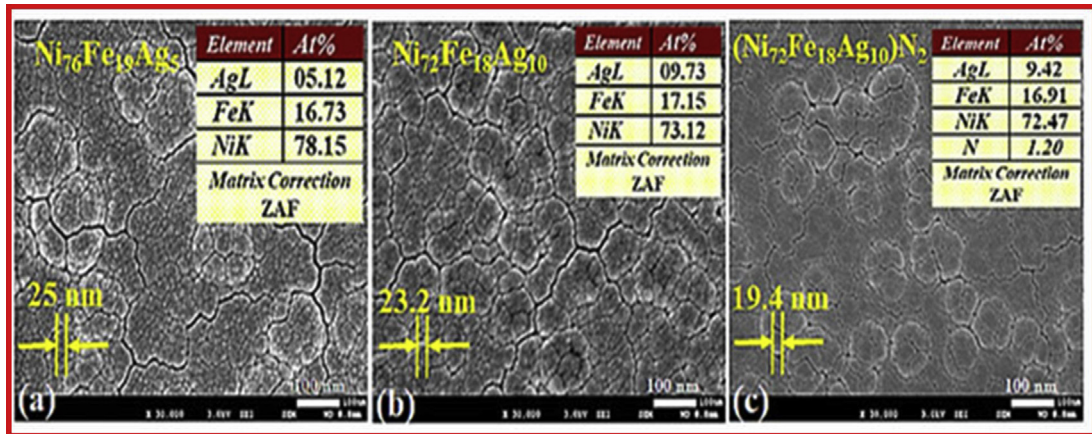


Figure 8.3

SEM images and EDS analysis of nanostructured silver doped permalloy thin films deposited at 115 W. (A) $\text{Ni}_{76}\text{Fe}_{19}\text{Ag}_5$; (B) $\text{Ni}_{72}\text{Fe}_{18}\text{Ag}_{10}$ and (C) $(\text{Ni}_{72}\text{Fe}_{18}\text{Ag}_{10})\text{N}_2$. Ar gas has been used for deposition of sample (A) and (B). For sample (C), Ar + N_2 gas has been used [35].

mixture. It shows that the grains have fine surface and their size is 19.4 nm. It is observed from the EDS analysis of Fig. 8.3C that nitrogen has mixed. The reactive gas (N_2) has modified the chemical composition of the film [35].

9. Applications

9.1 Electronic industries

Magnetron sputtering uses to produce the metallizing for microelectronic circuits and chip carriers, electrical resistance films (Ni—Cr for strain gauges). Magnetic films are another important area for magnetron sputtering. Magnetic thin films are used predominantly for read-write storage of data for computer equipment. Storage magnetic films (Co, Fe, Co—Pt, Co—Cr, Cu—Ni) for magnetic storage devices, floppy discs, tapes and thin-film magnetic heads, opto-storage devices (compact discs and video discs), corrosion-resistant films (Cr—Ni), bonding for interlayers, hermetic seals, brazing for lead and gas sensors [36,37].

Ion-beam sputtering has found application in the manufacture of thin-film heads for disk drives. Magnetron sputtering is also can be seen in computer hard disk, gate dielectric, thin-film transistors, low-emissivity coatings on glass, interlayer dielectric, on-chip interconnects and interlevel bias, passive thin film components, printed circuit boards, sensors, surface acoustic wave devices, double-pane window assemblies, super mirror as well [38,39].

**Figure 8.4**

(A) Sputter deposited CD for magnetic storage application, (B) thin-film coated sensor, (C) integrated circuits.

Sputtered NiTi film acts as an antivibration damping structure. It is responsible for minimizing the vibrations during the operations of the hard disk drive [40]. Sputtering is one of the main processes of manufacturing optical waveguides. It is applied for making of efficient photovoltaic solar cells [41]. Sputtering is also used as the process to deposit the metal (e.g., Al) layer during the fabrication of CDs and DVDs (Fig. 8.4A). Sputter deposited sensor and integrated circuit are given in Fig. 8.4B and C.

9.2 Automobile and aerospace industries

In automobile industries, the magnetron sputtering deposition can be seen in automobile wheels, rims, mirror, headlights and taillights, auto trim components, drive train bearings and components which proved to be a better surface resistance application. Auto car wheel, plastic/metal cutlery are used chrome magnetron sputtering of vacuum coating machine [42]. In aerospace industries, the magnetron sputtering deposition is useful for heads-up cockpit displays, jet turbine engines, mirrors for optical and X-ray telescopes, and night vision equipment [43]. Some of the magnetron sputtered automotive and aero-industries application have shown in Fig. 8.5.

9.3 Energy harvesting industries

In energy harvesting industries, sputtering deposition is utilized in gas turbine blade coatings, solar panels as well [44]. Though other processes are used for the production of the silicon, sputtering is widely used to produce a back reflector layer (made predominantly with Al) and front contacts (made with transparent conducting oxides) [45]. In addition to the traditional Al and transparent conducting oxides coatings, these technologies employ additional sputtered layers such as Mo and intrinsic ZnO [46].

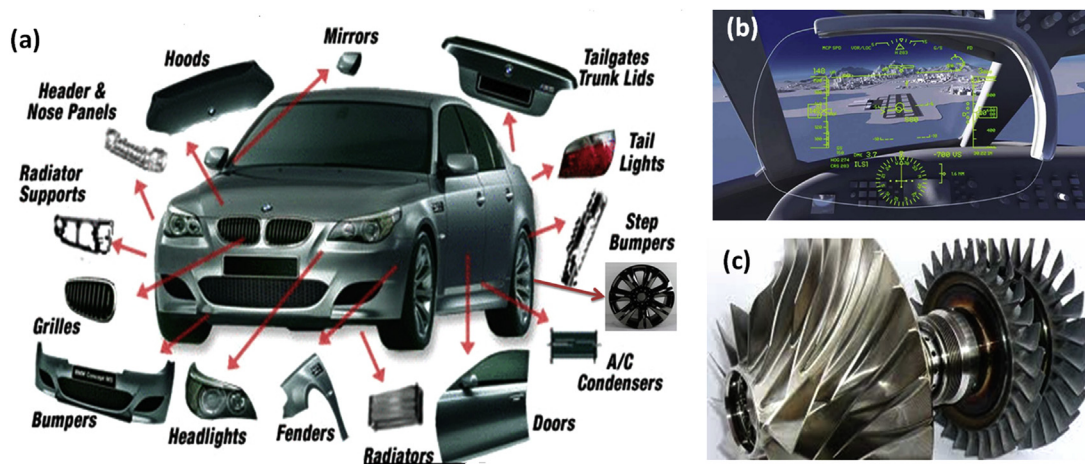


Figure 8.5

Sputter deposited (A) interior and exterior trim components in automobile, (B) heads-up cockpit displays of aircraft, and (C) surface area of aero-turbine engine.

Intrinsic ZnO can be produced using midfrequency AC sputtering at much higher rates than the traditional RF technology used with planar cathodes. The magnetron sputtering is also used to fabricate: (1) antireflective coatings (to allow to solar energy to pass through protective coverings such as glass), (2) reflective coatings for solar collectors, and (3) absorbing coatings to absorb the maximum solar radiation [47]. Fig. 8.6 shows the solar panel prepared by utilizing the magnetron sputtering deposition technique.

9.4 Biomedical industries

Generally, biomedical equipments are biocompatible, wear resistance, chemically inert, and very hard in nature. Magnetron sputtering is used to deposit diamond-like carbon coatings deposited on other materials in order to provide many of the beneficial properties

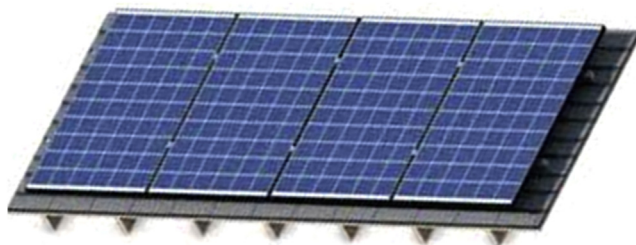


Figure 8.6

Magnetron sputter deposited solar panel.

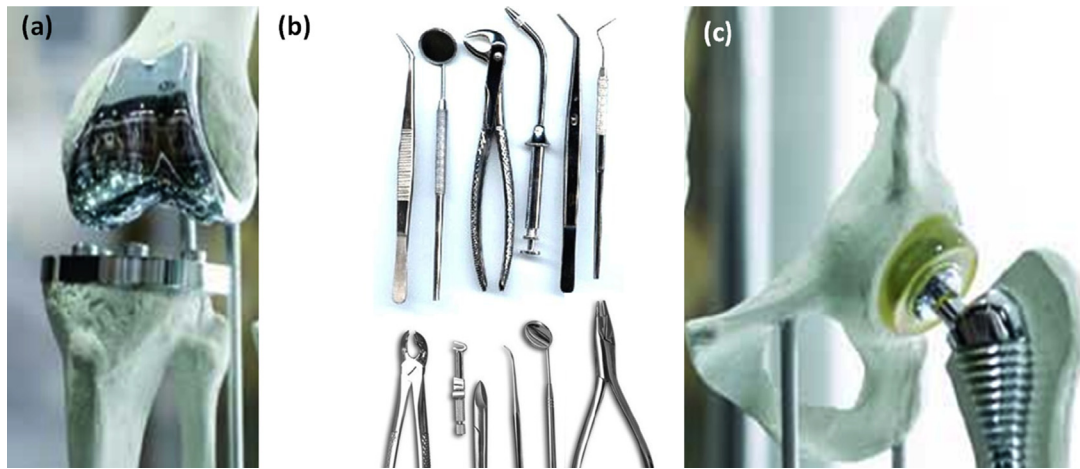


Figure 8.7

Hard surface deposition by RF magnetron sputtered (A) knee implant material, (B) surgical instruments, and (C) hip implant.

which minimizes friction [48]. Generally, Ti or Ti-based alloys and other biocompatible materials are used to sputter on the medical instruments [49]. Nitinol-smart material coated on the knee implant, hip implant, and surgical instrument using magnetron sputtering are shown in Fig. 8.7.

9.5 Textile industries

Magnetron sputtering technology can deposit metal or nonmetal films on the surface of textiles (substrate) such as polyester, cotton, linen, silk, wool, polyamide, polylactic acid, and polypropylene. This kind of film deposition is done by selecting appropriate sputtering process, different target materials and ambient gases [50]. The main structures of the textiles substrate include woven fabrics, knitted fabrics, and nonwoven fabrics. Here the sputtering target materials are metals (Cu, Ti, Ag, Al, W, Ni, Sn, Pt, etc.) or nonmetals (Si, graphite, etc.) as well as metal oxides (TiO_2 , Fe_2O_3 , WO_3 , ZnO etc.) and nonmetal oxide (SiO_2 , graphene oxide, etc.) [51,52].

Also, ceramic materials and single or multi-layer composite nanofilms formed with polymers (such as polyimide and polytetrafluoroethylene) can be deposited using this method. It have single or compound functions such as electromagnetic shielding, UV protection, antistatic, antibacterial, conductive, or waterproof properties [53]. A metal coated fibers and a heat resistance kitchen gloves prepared by magnetron sputtering deposition has been shown in Fig. 8.8.

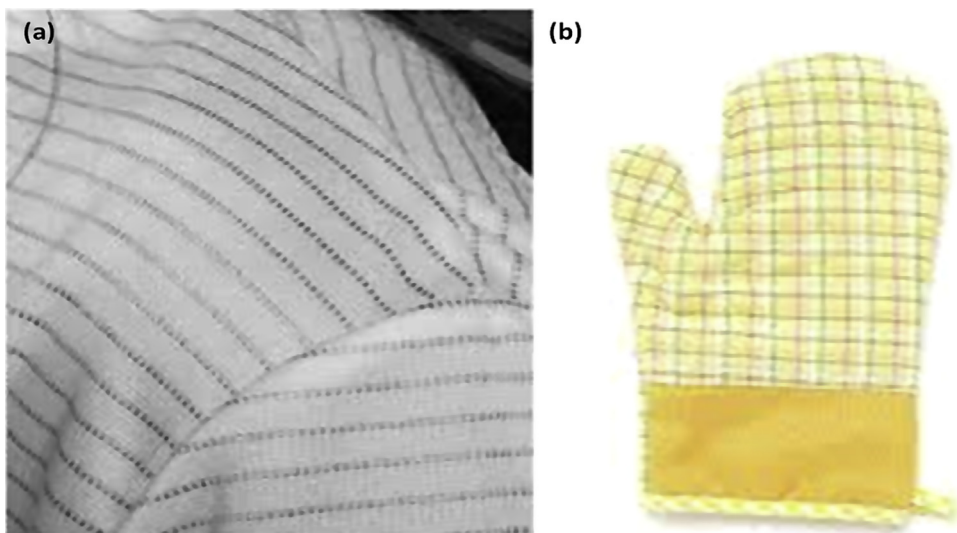


Figure 8.8

(A) Metal coated fibers by magnetron sputtering, (B) silicon sputtered heat resistant kitchen gloves.

9.6 Other industries

Sputtering technology spread a large area in glass coating either developing the transparent or opaque film. A large industry has developed around tool bit coating using sputtered nitrides, such as TiN, creating the familiar gold colored hard coat [54]. DC sputtering can be used for bonding films and other metallurgical films. Sputtered films of Cr, Ti, Ni, Cu, and Au have been applied successfully for the subsequent soldering and brazing treatments onto glass or ceramic substrates [55].

Superconducting films such as NbN, Si_3N_4 , and mixed oxide compositions can be conveniently prepared by RF sputtering [56]. Wear-resistant coatings can be deposited by RF magnetron sputtering from compound targets or by reactive sputtering. Material such as WC, Si_3N_4 , Al_2O_3 , and TiN can be sputtered to give very adherent hard coatings of thicknesses up to several micrometers [57]. Comparing to the CVD and MBE, magnetron sputtering is also an alternative method to grow GeSn alloys for optoelectronic applications [58].

A high-quality SiGeSn/GeSn multiple quantum well structure have also been grown on Ge substrate by sputtering epitaxy. In this structure, 6% Sn content of the GeSn layer mainly contributes for the functions of GeSn photo-detector [59]. Some sputtering applications used in optics industries are antireflective/antiglare coatings, cable communications, laser

lenses, optical filters for achromatic lenses, spectroscopy; wear resistance coatings: anticorrosion coatings, antiseize coatings, dies and molds, sewing needles, tool and drill bit hardening; in decorative: appliance trim, building glass, building hardware, clothing, jewelry, packaging, plumbing fixtures, and toys [60,61].

10. Limitations of magnetron sputtering

Sputtering is relatively simple, versatile process and most popular deposition technique. However, it has some limitations which are discussed below [62–64]:

- It contains the expensive targets, which limits the scope of their applications.
- A particular designed target is necessary to develop according to the sputtering machine to carry out the sputtering process.
- After few depositions, an erosion track is found on the target. For further deposition, it gives poor efficiency of sputtering yield.
- The target utilization can be increased by flattening the magnetic field lines parallel to the target surface.
- Most of the energy incident on the target becomes heat, which must be removed.
- In reactive sputter deposition, the gas composition must be carefully controlled to prevent poisoning the sputtering target.
- Compressive stresses may be generated in films during sputtering caused by the energetic species bombarding the substrate. In extreme cases, such stresses may cause the cracking and the delamination of the film from the substrate.
- Nonlinearity, complex thermomechanical behavior, ineffectiveness for precise and complex motion control, and force tracking.
- Sputtering is performed using a vacuum chamber in which the target(s) and the substrate are placed; thereafter the chamber is evacuated to vacuum.

Despite these excellent properties, full integration of smart alloy thin films into MEMS is limited. Their stoichiometry sensitivity has restricted the manufacturability and reliability of these materials in MEMS. The large exothermic heat of transformation limits the cycle lifetime. The poor machinability limits the many of its applications.

11. Influencing factors in magnetron sputtering

It is analyzed and observed from the various literatures that the magnetron sputtering depends on several factors such as angle/energy of the incident ions, total mass of the ions, binding energy of atom at the target, sputtering gas/power and substrate temperature/bias. Some of the factors are influencing the sputtering yield/output. They are enumerated below.

- Maintaining the higher density of ions improves the ionization process and sputter yield/deposition rate in magnetron sputtering.
- The ionized discharge can be increased by increasing the power and pressure.
- Increasing the microwave power and gas pressure increases the deposition rate in microwave amplified magnetron sputtering.
- The pressure of primary inert gas and reactive gas (O_2 and N_2) determines the chemical composition of oxide/nitride thin films produced in reactive magnetron sputtering process.
- Increased target-to-substrate distance leads to increase gas pressure.
- Increasing Ar gas pressure decreases the density and structural defects of the resultant thin films.
- Reducing the impurities (oxygen, hydrogen, and carbon) during smart material film formation improves shape memory effect and prevents brittleness and deterioration.
- Low conducting materials and insulator materials cannot be sputtered in DC magnetron sputtering due to nonpassing of the current in target materials and charge accumulation. Utilization of the RF magnetron sputtering and high frequency AC solve this limitation.
- Energy of ions and flux of ions can be controlled in ion-beam sputtering process.
- The RF power and substrate bias have influences on the ion flux/energy of the ions.
- High energized positive ions limit the maximum power in DC magnetron sputtering by thermal load on the target.
- High substrate temperature supports to yield denser and compact film.
- Applying of small negative bias voltage at the substrate end improves the quality of sputter deposited films.
- Low distance between sputtered material and the target attracts the metallic ions back to the target. It leads to the reduction in deposition rate (can be tuned by magnetic confinement).

12. Conclusion

This chapter has given a brief description on magnetron sputtering and its market demand over the manufacturing of thin films and nanostructured coatings. A distinct number of advantages give a unique position of sputtering deposition among all other deposition processes. It has been demonstrated that there are various magnetron sputtering techniques such as DC magnetron sputtering processes, RF magnetron sputtering, ion-beam magnetron sputtering, reactive sputtering, inductively coupled plasma-magnetron sputtering, microwave amplified magnetron sputtering, high power impulse magnetron sputtering, and high-target-utilization sputtering. NiTi smart materials fabrication using magnetron sputtering has been discussed. Various process parameters are associated practical problems are depicted briefly. Formation of nanocomposite using two different

way of magnetron sputtering has been discussed. A wide range of application from micro/nanoelectronics, smart structure to surface modifications of textile has been discussed with some of its process limitations.

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